

Sets of Cases (Enrichment)

Microarray experiments, through a combination of insufficient data per gene and the difficulties of large-scale simultaneous inference, often yield disappointing results. In search of greater detection power, *enrichment analysis* considers the combined outcomes of biologically determined sets of genes, for example the set of all the genes in a predefined genetic pathway. If all 20 z -values in a hypothetical pathway were positive, we might assign significance to the pathway's effect, whether or not any of the individual z_i were deemed non-null. We will consider enrichment methods in this chapter, and some of the theory, which of course applies just as well to similar situations outside the microarray context.

Our main example concerns the *p53 data*, partially illustrated in Figure 9.1; p53 is a transcription factor, that is, a gene that controls the activity of other genes. Mutations in p53 have been implicated in cancer development. A National Cancer Institute microarray study compared 33 mutated cell lines with 17 in which p53 was unmutated. There were $N = 10\,100$ gene expressions measured for each cell line, yielding a $10\,100 \times 50$ matrix X of expression measurements. Z -values based on two-sample t -tests were computed for each gene, as in (2.1)–(2.5), comparing mutated with unmutated cell lines. Figure 9.1 displays the 10 100 z_i values.

The results are disappointing. The histogram looks like a slightly short-tailed normal density, with MLE empirical null estimate $\mathcal{N}(0.06, 1.11^2)$. One gene, “BAX”, stands out at $z = 4.83$, but the other 10 099 genes have uninteresting fdr and Fdr values, even using a $\mathcal{N}(0, 1)$ null.

This is where enrichment analysis comes to at least a partial rescue. A collection of 522 potentially interesting gene sets were identified from pre-existing¹ catalogs relating to p53 function. The sets ranged in size from two to 358 genes, median 17, with considerable overlap among the sets.

¹ Here we are dealing only with pre-selected gene sets. “Unsupervised learning,” in which the sets are chosen on the basis of the observed data — say by a clustering algorithm — lies beyond the scope of the theory.

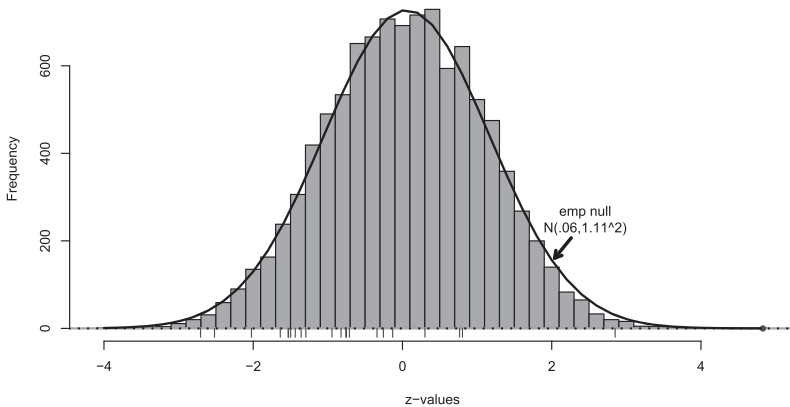


Figure 9.1 Histogram of $N = 10\,100$ z -values from the p53 study. Curve shows the empirical null density $\mathcal{N}(0.06, 1.11^2)$. False discovery rate tests label all genes null, except for BAX gene at $z = 4.83$. Enrichment analysis, however, will yield 8 non-null gene sets. Hash marks indicate the 22 z -values for gene set *ras pathway*.

We will see that eight of the 522 sets indicate non-null enrichment, even taking account of having 522 candidate sets to consider. One of the eight is the *ras pathway*, whose 22 z_i values, indicated by hash marks in Figure 9.1, are mostly < 0 .

9.1 Randomization and Permutation

We have an $N \times n$ data matrix X and a corresponding N -vector of scores z , with z_i determined from the i th row of X . The p53 study, with $N = 10\,100$, takes z_i to be the two-sample statistic (2.1)–(2.5), but our calculations will not require any specific form for the z -values. To begin with, we consider a single gene set² $\mathcal{S}$ comprising m z -values,

$$z_{\mathcal{S}} = \{z_i, i \in \mathcal{S}\} \quad [\#\mathcal{S} = m]; \quad (9.1)$$

$m = 22$ for the *ras pathway* in Figure 9.1. Our goal is to assign a p -value $p_{\mathcal{S}}$ to the null hypothesis H_0 of “no enrichment” for $\mathcal{S}$. Later we will use false discovery rate methods to take into account simultaneous testing on a catalog of possible gene sets.

² Microarray terminology is convenient here, but of course the theory and methods apply more generally.

Let $S(z_S)$ be a score we attach to the z -values in S . We will mainly deal with simple averages,

$$S(z_S) = \sum_{i \in S} s(z_i)/m \quad (9.2)$$

where $s(z)$ is some function like z or z^2 , but more complicated scoring functions are also of interest. A popular scoring choice is the *GSEA statistic* (gene set enrichment analysis), essentially the maximum absolute difference³ between the cumulative distribution functions of the m z -values in S and the $N - m$ z -values not in S . A p -value p_S is obtained by calculating resampled versions of S and seeing what proportion of them exceed $S(z_S)$.

The problem is that there are two quite plausible but different resampling schemes:

Column permutations The columns of X are randomly permuted giving recalculated z -value vectors $z^{*1}, z^{*2}, \dots, z^{*B}$ as in Section 6.5, corresponding scores

$$S^{*b} = S(z^{*b}) \quad b = 1, 2, \dots, B, \quad (9.3)$$

and p -value

$$p_S^* = \# \{S^{*b} > S\} / B. \quad (9.4)$$

Row randomization Random subsets of size m are selected from $\{1, 2, \dots, N\}$, say $i^{\dagger 1}, i^{\dagger 2}, \dots, i^{\dagger B}$, giving corresponding m -vectors and scores

$$z^{\dagger b} = \{z_i, i \in i^{\dagger b}\}, \quad S^{\dagger b} = S(z^{\dagger b}), \quad (9.5)$$

and p -value

$$p_S^\dagger = \# \{S^{\dagger b} > S\} / B. \quad (9.6)$$

Exercise 9.1 In what way does a single permutation z -vector z^* require more computation than a single randomization vector $z^\dagger$?

Row randomization has the appealing feature of operating *conditionally* on the observed vector z : it tests the null hypothesis that S is no bigger than what might be obtained by a random selection of m elements from z .

Its defect concerns correlation between the genes. Suppose the N values $s_i = s(z_i)$ in (9.2) have empirical mean and standard deviation

$$s \sim (m_s, \text{sd}_s). \quad (9.7)$$

³ This difference is a version of the Kolmogorov–Smirnov test statistic.

Then a row-randomized version of (9.2),

$$S^\dagger = \sum_{i \in S^\dagger} s(z_i)/m \quad (9.8)$$

will have mean and standard deviation

$$S^\dagger \sim (\mu^\dagger, \sigma^\dagger) = (m_s, \text{sd}_s / \sqrt{m}) \quad (9.9)$$

just as if the genes were independent of each other.

Exercise 9.2 Formula (9.9) correctly gives $(\mu^\dagger, \sigma^\dagger)$ if we allow the subsets $i^\dagger$ to be chosen randomly and with replacement from $\{1, 2, \dots, N\}$. What if we insist that they be drawn *without* replacement?

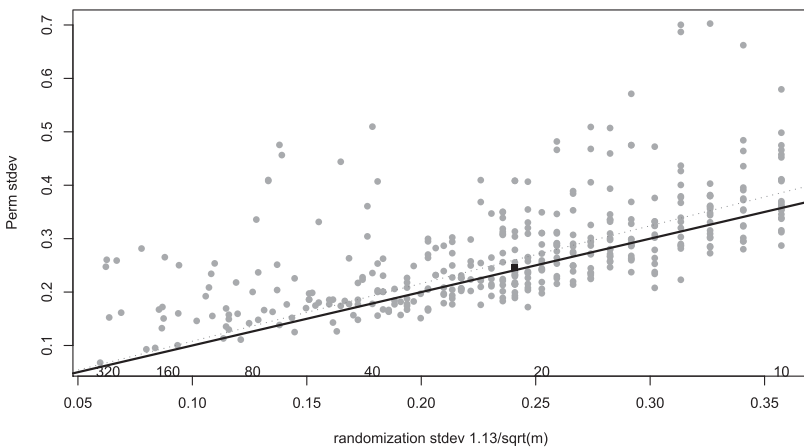


Figure 9.2 Permutation standard deviations compared to randomization standard deviations for $S = \bar{z}_S$, the average z -value; 395 gene sets with $m \geq 10$, p53 data. The points tend to lie above the main diagonal (heavy line), especially for large m (m indicated by numbers above x -axis). Dark square is *ras* pathway.

Taking $s(z_i) = z_i$ for the p53 data gives mean and standard deviation⁴

$$s \sim (0.00, 1.13). \quad (9.10)$$

Figure 9.2 compares $\sigma_S^\dagger = 1.13 / \sqrt{m}$ with the column permutation standard

⁴ The catalog of 522 gene sets includes, with repetitions, 15 059 genes, of which 4486 are distinct. The values in (9.10) are those for the list of 15 059 rather than (0.04, 1.06) for the original 10 100 values; (9.10) seems more appropriate for the comparison made in Figure 9.2.

deviation σ_S^* obtained from $B = 1000$ values S^{*b} (9.3) (for the 395 gene sets $\mathcal{S}$ having $m \geq 10$ members). The ratio $\sigma_S^*/\sigma_S^\dagger$ has median 1.08 (light line) but is considerably greater for large m , with median 1.50 for sets $\mathcal{S}$ with $m \geq 50$.

Positive correlations between the gene expressions in $\mathcal{S}$ make the permutation standard deviation σ_S^* of S greater than the row randomization value $\sigma_s = \text{sd}_s / \sqrt{m}$. The randomization p -value (9.6) will then be misleadingly small, at least compared with p_S^* , the usual permutation value.

Exercise 9.3 Let s_S^* be the m -vector of values $s(z_i^*)$, $i \in \mathcal{S}$, corresponding to a permutation vector z_S^* , as in (9.3). If Σ_S is the $m \times m$ covariance matrix of s_S^* under permutation, show that $\sigma_S^*/\sigma_S^\dagger = \bar{\Sigma}_S^{1/2} / \text{sd}_s$, with $\bar{\Sigma}_S$ the average of the m^2 elements of Σ_S . (So $\Sigma_S = I$ gives $\sigma_S^* = \sigma_S^\dagger / \text{sd}_s$, but positive off-diagonal elements make $\sigma_S^* > \sigma_S^\dagger / \text{sd}_s$.)

Permutation tests for enrichment have a serious weakness of their own, as demonstrated by the following example. The BRCA microarray data of Section 8.4 involved $n = 15$ breast cancer patients, seven with BRCA1 mutations, eight with BRCA2, on $N = 3226$ genes; z -values for each gene were calculated as in (2.1)–(2.5), comparing the two BRCA categories. The z -value histogram was highly overdispersed, empirical null estimate $\mathcal{N}(-0.04, 1.52^2)$; Fdr analysis yielded no non-null genes.

An artificial catalog of 129 gene sets $\mathcal{S}$, each of size $m = 25$, was constructed by random selection from all N genes, making it highly unlikely that any set was actually enriched. Enrichment was tested using $S = \sum_S |z_i|/m$, that is, (9.2) with $s(z) = |z|$. Figure 9.3 shows that the permutation values S^* greatly underestimated the actual 129 S values. The permutation p -values (9.4), $B = 400$, were very small for most of the sets $\mathcal{S}$: a standard Fdr analysis (BH(0.1) as in Section 4.2) labeled 113 of the 129 as “enriched,” even though we know that all of them were chosen at random.

It is easy to spot the trouble here: the mean and standard deviation of the $N = 3226$ values $|z_i|$ is

$$(m_s, \text{sd}_s) = (1.17, 0.82) \quad (9.11)$$

compared to the mean and standard deviation (m_s^*, sd_s^*) for all $N \cdot B$ permutation values $|z_i^{*b}|$,

$$(m_s^*, \text{sd}_s^*) = (0.81, 0.59). \quad (9.12)$$

This makes a typical permuted score $S^* = \sum_S |z_i^*|/m$ smaller than the actual score $S = \sum_S |z_i|/m$.

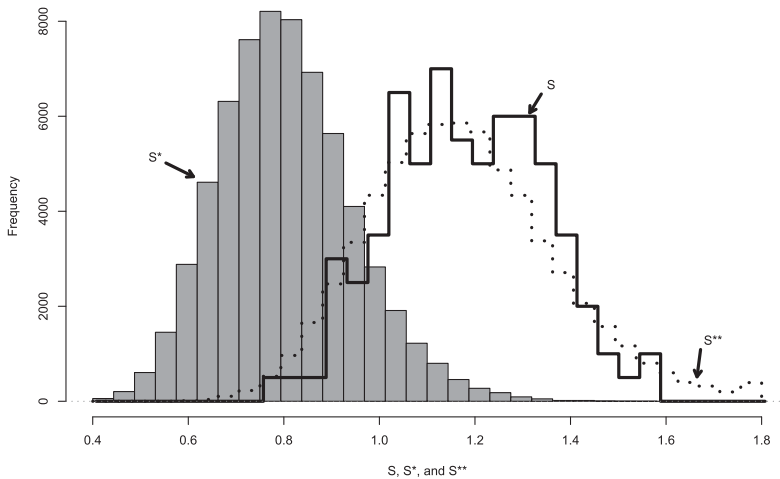


Figure 9.3 Line histogram shows scores $S = \sum_S |z_i|/m$ for 129 randomly selected gene sets from BRCA data, all $m = 25$. Solid histogram is permutation scores $S^{*b} = \sum_S |z_i^{*b}|/m$, $b = 1, 2, \dots, 400$. Dotted histogram shows restandardized scores S^{**} (9.13).

The values reported in (9.12) are just about what we would get if the z_i^* replicates followed a theoretical $\mathcal{N}(0, 1)$ null distribution. This is a dependable feature of permutation t -tests, Section 6.5, which are immune to the overdispersion seen in the BRCA z -values. The actual scores $S = \sum_S |z_i|/m$ are not immune, and are shifted to the right in Figure 9.3.

Row randomization handles overdispersion correctly, but ignores correlation; column permutation ignores overdispersion but correctly accounts for correlation. *Restandardization* is a somewhat ad hoc attempt to deal correctly with both problems. Given a permutation score S^{*b} (9.3), we compute the *restandardized score*

$$S^{**b} = m_s + \frac{\text{sd}_s}{\text{sd}_s^*} (S^{*b} - m_s^*) \quad (9.13)$$

where (m_s, sd_s) are from (9.7) while (m_s^*, sd_s^*) are the permutation mean and standard deviation of $s(z_i^{*b})$ over all B permutations and N genes as in (9.13). The restandardized p -value is then

$$p_S^{**} = \# \{S^{**b} > S\} / B. \quad (9.14)$$

The dotted curve in Figure 9.3 is the histogram of all $129 \cdot B$ restandardized scores S^{**} . It reasonably matches the actual S histogram; a BH(0.1)

analysis of the 129 p_S^{**} values now correctly declares all the sets to be un-enriched.

Another way to write (9.14) is

$$p_S^{**} = \# \left\{ \frac{S^{*b} - m_s^*}{sd_s^*} > \frac{S - m_s}{sd_s} \right\} / B. \quad (9.15)$$

If we change $s(z)$ to the *standardized* function

$$t(z) = (s(z) - m_s) / sd_s \quad (9.16)$$

with (m_s, sd_s) thought of as fixed constants, e.g., as in (9.11), then the corresponding scoring function is

$$T(z) = \sum_S t(z_i) / m = (S(z) - m_s) / sd_s \quad (9.17)$$

and (9.15) becomes

$$p_S^{**} = \# \{ T^{*b} > T \} / B. \quad (9.18)$$

In this sense, restandardization amounts to calculating the usual permutation p -value for a row-standardized version of S .

Exercise 9.4 Verify (9.18). *Hint:* $(m_t, sd_t) = (0, 1)$.

The need for restandardization resembles the arguments for the empirical null in Chapter 6. Theoretically (9.11) should match (9.12) in the BRCA example, but factors like those of Section 6.4 have spread out the distribution of $|z_i|$. These factors don't affect the permutation distribution, and some form of standardization is necessary before we can trust permutation p -values. This doesn't have to be of form (3.16). The GSEA algorithm achieves a similar effect by comparing the cdf of the z -values in S to the cdf of all the others rather than to a theoretical $\mathcal{N}(0, 1)$ cdf.

Restandardization can be shown to yield reasonable inferences in a variety of situations:

- if S was selected randomly, as in the row randomization model;
- if the theoretical null $z \sim \mathcal{N}(0, 1)$ agrees with the empirical distribution of the N z -values;
- if the z_i are uncorrelated.

The method is not perfect, as examples can be constructed to show. Neither the randomization nor permutation models perfectly describes how gene sets S come under consideration in practice, making some sort of compromise formula a necessity. Section 9.3 provides some theoretical support for formula (9.13), in an admittedly specialized formulation.

9.2 Efficient Choice of a Scoring Function

The previous section concerned the proper choice of a null hypothesis for enrichment testing, with the control of Type I error (false rejection) in mind. We are also interested in power, of course. This requires us to specify alternatives to null selection, which then leads to recipes for efficient scoring functions $S(z)$.

Row randomization (9.6) tests the null hypothesis that set $\mathcal{S}$ was chosen by random selection from the N cases. The *Poisson selection model* allows non-random selection. It starts with independent Poisson variates

$$I_i \stackrel{\text{ind}}{\sim} \text{Poi}(v_i) \quad v_i = \alpha e^{\beta' s_i} / \tau_\beta \quad (9.19)$$

for $i = 1, 2, \dots, N$, where $s_i = s(z_i)$ is some *feature function*, a J -dimensional function that we believe extracts the features of z_i indicative of enrichment, e.g., $s(z) = (z, z^2)'$; β is an unknown J -dimensional parameter vector, and

$$\tau_\beta = \sum_{i=1}^N e^{\beta' s_i}; \quad (9.20)$$

α is an unknown scalar parameter satisfying

$$\alpha = \sum_{i=1}^N v_i \quad (9.21)$$

according to (9.19)–(9.20).

In what follows, v_i will be small and the I_i almost all zeroes or ones, mostly zeroes. It will be notationally convenient to assume that they are *all* 0 or 1. Then we define the selected gene set $\mathcal{S}$ as composed of those cases having $I_i = 1$,

$$\mathcal{S} = \{i : I_i = 1\} \quad (9.22)$$

so $\mathcal{S}$ has

$$m = \sum_1^N I_i \quad (9.23)$$

members. The Poisson assumptions (9.19) give $m \sim \text{Poi}(\sum v_i)$, or by (9.21),

$$m \sim \text{Poi}(\alpha). \quad (9.24)$$

It is easy to calculate the probability $g_{\alpha, \beta}(\mathcal{S})$ of selecting any particular gene set $\mathcal{S}$ under model (9.19),

$$g_{\alpha, \beta}(\mathcal{S}) = \left(\frac{e^{-\alpha} \alpha^m}{m!} \right) \left(m! e^{m[\beta' \bar{s}_{\mathcal{S}} - \log(\tau_\beta)]} \right) \quad (9.25)$$

where

$$\bar{s}_S = \sum_S s_i/m, \quad (9.26)$$

called $S(z_S)$ in Section 9.1. This is a product of two exponential family likelihoods, yielding maximum likelihood estimates $\hat{\alpha} = m$ and $\hat{\beta}$ satisfying

$$\sum_{i=1}^N s_i e^{\hat{\beta}' s_i} \bigg/ \sum_{i=1}^N e^{\hat{\beta}' s_i} = \bar{s}_S. \quad (9.27)$$

Exercise 9.5 Verify (9.25) and (9.27). Show that the conditional distribution of S given its size m is

$$\begin{aligned} g_\beta(S|m) &= m! e^{m[\beta' \bar{s}_S - \log(\tau_\beta)]} \\ &= m! e^{m[\beta'(\bar{s}_S - \bar{s}) - (\log(\tau_\beta) - \beta' \bar{s})]} \end{aligned} \quad (9.28)$$

where $\bar{s} = \sum_1^N s_i/N$.

Parameter vector $\beta = \mathbf{0}$ corresponds to the row randomization null hypothesis that S was selected at random. Non-zero β “tilts” the selection toward gene sets having larger values of $\beta'(\bar{s}_S - \bar{s})$, as seen in (9.28). An efficient test rejects $H_0 : \beta = 0$, i.e., that S was randomly selected, if $\bar{s}_S$ is far from its null expectation $\bar{s}$. If $s(z)$ is one-dimensional, as in Section 9.1, we reject for extreme values of $\bar{s}_S - \bar{s}$, either positive or negative, perhaps using permutation methods to decide what “far from” means.

The choice of $s(z)$ determines which alternatives the test will have power against. Consider two candidates,

$$s(z) = z \quad \text{or} \quad s(z) = |z|. \quad (9.29)$$

The first of these rejects if $\bar{z}_S$ is far from $\bar{z} = \sum_1^N z_i/N$: a good choice if enrichment manifests itself as a *shift* of location in z -values, as seems to be the case with the *ras pathway* in Figure 9.1. The second candidate has power against *scale* alternatives, where the z -values in S are more dispersed than in the whole population.

For general use one would like to have power against both shift and scale alternatives, which suggests using a two-dimensional statistic $s(z)$. The *maxmean statistic* takes

$$s(z) = (s^{(1)}(z), s^{(2)}(z)) \quad \begin{cases} s^{(1)}(z) = \max(z, 0) \\ s^{(2)}(z) = -\min(z, 0) \end{cases} \quad (9.30)$$

giving bivariate score function $S(z_S) = (\bar{s}_S^{(1)}, \bar{s}_S^{(2)})$. This is reduced to a

one-dimensional test statistic by maximization,

$$S_{\max\text{mean}} = \max(\bar{s}_S^{(1)}, \bar{s}_S^{(2)}). \quad (9.31)$$

(Notice that this is not the same as $S_{\text{absmean}} = \max(\bar{z}_S, -\bar{z}_S) = |\bar{z}_S|$, the two-sided version of the test based on means.)

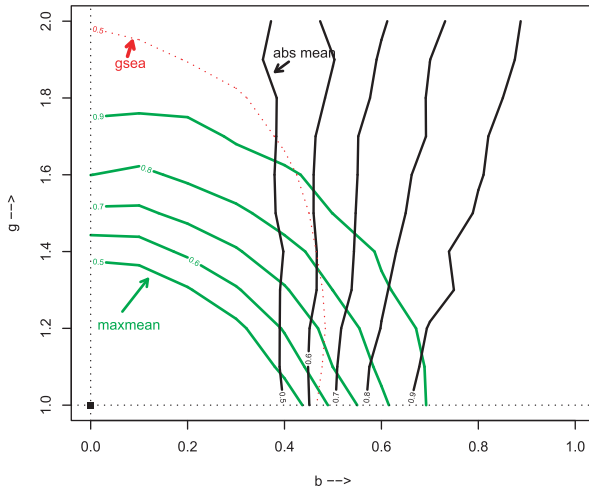


Figure 9.4 Contours of equal power for testing $(b, g) = (0, 1)$ in situation (9.32); for $S_{\max\text{mean}}$ (9.31), GSEA (Kolmogorov–Smirnov) statistic, and $S_{\text{absmean}} = |\bar{z}_S|$. Comparing contours for power = 0.5 shows $S_{\max\text{mean}}$ dominating GSEA and nearly dominating S_{absmean} .

Figure 9.4 compares the power of the maxmean test with that of the GSEA (Kolmogorov–Smirnov) statistic and also with $S_{\text{absmean}} = |\bar{z}_S|$ in an artificially simplified context: S consists of $m = 25$ independent normal observations,

$$z_i \stackrel{\text{ind}}{\sim} \mathcal{N}(b, g^2) \quad i = 1, 2, \dots, m = 25. \quad (9.32)$$

Contours of equal power are shown for testing

$$H_0 : (b, g) = (0, 1) \quad (9.33)$$

versus alternatives $(b \geq 0, g \geq 1)$ at level 0.95.

Better power is indicated by contours closer to the null hypothesis point $(b, g) = (0, 1)$. The maxmean test shows reasonable power in both directions. Comparing contours for power = 0.5 shows maxmean dominating

GSEA and nearly dominating the absolute mean test.⁵ (The vertically oriented contours for S_{absmean} show that, as expected, it has power against shift alternatives ($b > 0$) but not scale alternatives ($s > 1$).)

Enrichment testing was carried out using S_{maxmean} on the catalog of 522 gene sets S ; $B = 1000$ restandardized values yielded p -values p_S^{**} for each set. A Benjamini–Hochberg Fdr test with control level $q = 0.1$ (4.9)–(4.10) yielded eight non-null cases, as illustrated in Figure 9.5. The top seven show shifts of their z -values, five right and two left, and would have been discovered by S_{absmean} but the eighth set shows more of a scale effect. Note that restandardization formula (9.13) was applied separately to $s^{(1)}$ and $s^{(2)}$ in (9.30) and then combined to give

$$S_{\text{maxmean}}^{**} = \max \left\{ \bar{s}_S^{(1)**}, \bar{s}_S^{(2)**} \right\}. \quad (9.34)$$

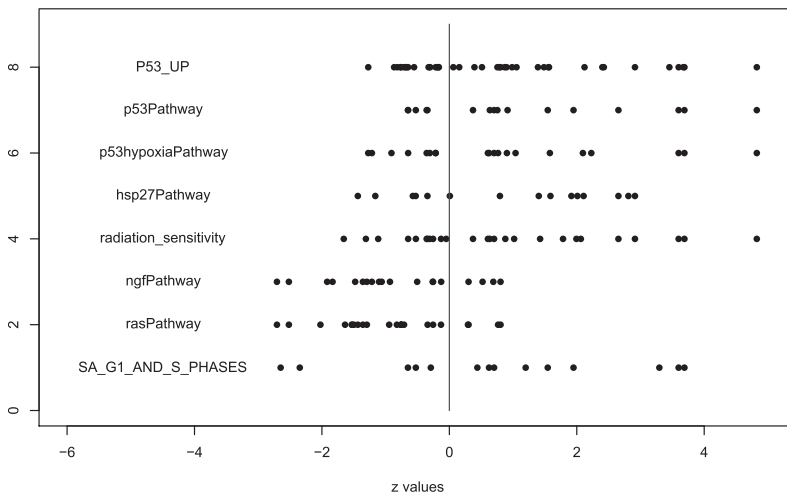


Figure 9.5 z -values for the 8 enriched gene sets, p53 data; non-null cases from catalog of 522 sets, Benjamini–Hochberg FDR control algorithm, $q = 0.10$, using p -values from maxmean statistic.

⁵ The power calculations were done by simulation from model (9.32), for example comparing the distribution of S_{maxmean} for $(b, g) = (0, 1)$ with the distribution for a grid of alternatives ($b \geq 0, g \geq 1$). In this context, the GSEA statistic was essentially the Kolmogorov–Smirnov distance between the cdf of $z_1, z_2, \dots, z_{25}$ and a standard normal cdf.

A word about computation: With 522 gene sets to consider, $B = 1000$ resamples is not excessive. The eighth largest p -value needs to be less than 0.00153 in order to be declared non-null by the BH(0.1) procedure; that is, less than two in 1000 values of S_S^{**} can exceed S_S . Calculations were carried out using the R program GSA, available through the CRAN library. It would be convenient to pool the resamples for different gene sets. This is the usual procedure for the permutation analysis of individual genes, for example in the SAM procedure of Section 4.5. However, the gene set resampling distributions depend heavily on both the size and correlation structure of S , making pooling dangerous.

Exercise 9.6 Verify the threshold value 0.00153 above.

The Poisson selection model (9.19)–(9.22) follows row randomization (9.5) in operating *conditionally* on the full collection z of N original z -values, with only the selection process for S considered random. We could instead think of S as fixed and the set of z -values in S , z_S , as random, perhaps following an exponential family of densities

$$f_\beta(z_S) = c_\beta e^{\beta S(z_S)} f_0(z_S). \quad (9.35)$$

Here $f_0(z_S)$ is the baseline density for an unenriched set, β is an unknown parameter, c_β is a constant making f_β integrate to 1, and S is some scoring function that produces larger values under enrichment.

In this case the optimum test for null hypothesis $H_0 : \beta = 0$ rejects for $S(z_S)$ large. By choosing S as the GSEA statistic, GSEA becomes optimum, etc., and we might very well use permutation/randomization methods to decide what “large” means. Taking $S = \sum_S s(z_i)/m$ gets us back to our previous theory, still leaving us with the task of deciding on the appropriate form for $s(z)$.

9.3 A Correlation Model

Restandardization attempts to correct enrichment p -values for both permutation and randomization effects (9.13)–(9.14). This section⁶ discusses a specialized model for correlated z -values where exact calculations are possible. We will see that restandardization fails to give exact answers, but is reasonably accurate over a range of situations. (Arguments and proofs are deferred to the section’s end.)

⁶ The material of this section is technical in nature and may be bypassed at first reading.

Our model assumes that the $N \times n$ data matrix $\mathbf{X}$ is of the form

$$\mathbf{X} = \mathbf{U} + \mathbf{V} \mathbf{a} \quad (9.36)$$

$N \times n \quad N \times n \quad N \times J \quad J \times n$

where $\mathbf{U}$ and $\mathbf{a}$ have mutually independent normal entries,

$$\begin{aligned} u_{ik} &\stackrel{\text{ind}}{\sim} \mathcal{N}(0, 1) & i = 1, 2, \dots, N & \text{ and } k = 1, 2, \dots, n \\ a_{jk} &\stackrel{\text{ind}}{\sim} \mathcal{N}(0, \sigma_a^2) & j = 1, 2, \dots, J & \text{ and } k = 1, 2, \dots, n; \end{aligned} \quad (9.37)$$

σ_a^2 is assumed known in what follows. J is a small integer dividing N , say with

$$H \equiv N/J \quad \text{and} \quad J_1 \equiv J - 1. \quad (9.38)$$

$\mathbf{V}$ is a fixed $N \times J$ matrix,

$$\mathbf{V}_{N \times J} = \begin{pmatrix} 1_H & -1_H/J_1 & \cdots \\ -1_H/J_1 & 1_H & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \quad (9.39)$$

where 1_H is a vector of H 1's. (All the off-diagonal vectors in (9.39) are $-1_H/J_1$.)

Exercise 9.7 Let $\mathbf{I}_J$ be the $J \times J$ identity matrix and $\mathbf{E}_H$ the $H \times H$ matrix of 1's. Show that

$$\mathbf{V}'_{J \times J} \mathbf{V} = N \frac{J}{J_1^2} (\mathbf{I}_J - \mathbf{E}_J/J) \quad (9.40)$$

and

$$\mathbf{V}'_{N \times N} \mathbf{V} = \frac{J}{J_1} \begin{pmatrix} \mathbf{E}_H & -\mathbf{E}_H/J_1 & \cdots \\ -\mathbf{E}_H/J_1 & \mathbf{E}_H & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \quad (9.41)$$

We assume that n is even, and define our vector of N z -values in terms of the difference of averages of the last and first $n/2$ columns of $\mathbf{X}$, say $\bar{\mathbf{x}}_2$ and $\bar{\mathbf{x}}_1$,

$$\mathbf{z} = \sqrt{n/4} (\bar{\mathbf{x}}_2 - \bar{\mathbf{x}}_1) / \sigma_0 \quad (9.42)$$

with

$$\sigma_0^2 = 1 + \frac{J}{J_1} \sigma_a^2. \quad (9.43)$$

(z_i is a two-sample t -statistic for case i , with the denominator known rather than estimated.) In terms of (9.36),

$$\mathbf{z} = (\mathbf{u} + \mathbf{V}\mathbf{A})/\sigma_0 \quad (9.44)$$

where, in notation similar to (9.42), the independent vectors $\mathbf{u}$ and $\mathbf{A}$ are defined as

$$\begin{aligned} \mathbf{u} &= \sqrt{n/4}(\bar{U}_2 - \bar{U}_1) \sim \mathcal{N}_N(\mathbf{0}, \mathbf{I}_N), \\ \mathbf{A} &= \sqrt{n/4}(\bar{a}_2 - \bar{a}_1) \sim \mathcal{N}_J(\mathbf{0}, \sigma_a^2 \mathbf{I}_J); \end{aligned} \quad (9.45)$$

(9.44)–(9.45) imply that

$$\mathbf{z} \sim \mathcal{N}_N(\mathbf{0}, (\mathbf{I}_N + \sigma_a^2 \mathbf{V}\mathbf{V}')/\sigma_0^2). \quad (9.46)$$

We will use $\mathbf{z}$ to illustrate correlation effects on the permutation and randomization calculations of Section 9.1. Formula (9.41) shows that the components z_i are $\mathcal{N}(0, 1)$ with correlation between z_{i_1} and z_{i_2} depending on whether i_1 and i_2 are in the same H -dimensional block of $\mathbf{V}$ in (9.39),

$$\begin{pmatrix} z_{i_1} \\ z_{i_2} \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho_{i_1, i_2} \\ \rho_{i_1, i_2} & 1 \end{pmatrix}\right)$$

where

$$\rho_{i_1, i_2} = \begin{cases} \rho & \text{same block} \\ -\rho/J_1 & \text{different blocks} \end{cases} \quad \left[\rho = \frac{\sigma_a^2}{J_1/J + \sigma_a^2} \right]. \quad (9.47)$$

The root mean square correlation α of Section 8.2 is nearly⁷

$$\alpha \doteq \frac{1}{\sqrt{J_1}}\rho \quad \left[\sigma_a^2 = \frac{J_1}{J} \frac{\sqrt{J_1}\alpha}{1 - \sqrt{J_1}\alpha} \right]. \quad (9.48)$$

Given a gene set $\mathcal{S}$ of size m , let $\bar{z}_{\mathcal{S}}$ be the average z -value in $\mathcal{S}$,

$$\bar{z}_{\mathcal{S}} = \sum_{\mathcal{S}} z_i / m. \quad (9.49)$$

Our aim is to show the role of restandardization, both its benefits and limitations, in assessing the variance of $\bar{z}_{\mathcal{S}}$. To this end, let

$$m_j = \#\{\text{members of } \mathcal{S} \text{ in } j\text{th block of } V\}, \quad (9.50)$$

⁷ The only approximation is setting $\rho_{i_1 i_2} = \rho$ instead of 1, so (9.48) is accurate to order $O(1/N)$.

$j = 1, 2, \dots, J$, that is, the number of z_i values in $\mathcal{S}$ having $(j-1)H < i \leq jH$ (so $\sum m_j = m$). Define

$$C = \frac{J}{J_1} \sum_{j=1}^J (m_j - m/J)^2 / m. \quad (9.51)$$

Lemma 9.1 *The marginal distribution of $\bar{z}_S$ averaging over both $\mathbf{u}$ and $\mathbf{A}$ in (9.44)–(9.45) is*

$$\bar{z}_S \sim \mathcal{N}\left(0, \frac{1}{m} \frac{1 + (J/J_1)\sigma_a^2 C}{1 + (J/J_1)\sigma_a^2}\right). \quad (9.52)$$

(Proof near the end of this section.)

C is a measure of correlation within $\mathcal{S}$. If all m members of $\mathcal{S}$ are in the same block, then the correlations (9.47) within $\mathcal{S}$ are big, giving $C = m$. At the other extreme, if $\mathcal{S}$ is perfectly balanced across blocks, $m_j \equiv m/J$, then most correlations are small, and $C = 0$.

Exercise 9.8 Show that if $\mathcal{S}$ is randomly selected by m draws without replacement from all N genes then

$$E\{C\} = 1. \quad (9.53)$$

C is connected with permutation calculations, since these correctly account for correlations within $\mathcal{S}$. We will see that row randomization effects are connected with

$$\hat{\sigma}_A^2 = \sum_{j=1}^J (A_j - \bar{A})^2 / J_1 \quad \left(\bar{A} = \sum_1^J A_j / J \right), \quad (9.54)$$

the sample variance of the random effects A_j in (9.44)–(9.45).

Theorem 9.2 *Conditionally on $\hat{\sigma}_A^2$, $\bar{z}_S$ has mean 0 and variance*

$$\text{var} \left\{ \bar{z}_S \middle| \hat{\sigma}_A^2 \right\} = \frac{1}{m} \frac{1 + (J/J_1)\hat{\sigma}_a^2 C}{1 + (J/J_1)\sigma_a^2}. \quad (9.55)$$

(Proof near the end of this section.)

Large values of $\hat{\sigma}_A^2$ make the blocks of z -values from (9.44) more disparate, and increase the variability (9.5) of $\bar{z}_{S^\dagger}$ for randomly selected subsets $\mathcal{S}^\dagger$. With $s(z) = z$ in (9.7), the component variance

$$\text{sd}_s^2 = \|z^2\| / N \quad (9.56)$$

is conditionally distributed as a scaled non-central χ^2 variate,

$$\text{sd}_s^2 | \hat{\sigma}_A^2 \sim \chi_N^2(\delta^2) / (N\sigma_0^2) \quad [\delta^2 \equiv N(J/J_1)\hat{\sigma}_A^2] \quad (9.57)$$

as verified at the end of this section. (All the means are known to be zero in model (9.44) so we are taking $m = 0$ in (9.7).) Familiar χ^2 properties yield

$$\text{sd}_s^2 | \hat{\sigma}_A^2 = \frac{1 + (J/J_1)\hat{\sigma}_A^2}{1 + (J/J_1)\sigma_a^2} + O_p\left(N^{-\frac{1}{2}}\right). \quad (9.58)$$

This means that the factor $\hat{\sigma}_A^2$ in (9.55) is effectively recoverable from sd_s (9.56).

The unconditional distribution of $\hat{\sigma}_A^2$ in (9.54) is a scaled central χ^2 ,

$$\hat{\sigma}_A^2 \sim \sigma_a^2 \chi_{J_1}^2 / J_1 \quad (9.59)$$

according to (9.45); $\hat{\sigma}_A^2$ has coefficient of variation $(2/J_1)^{1/2}$. Taking $J = 5$, as at the end of Section 8.2, gives $\text{CV} = 0.707$ so $\hat{\sigma}_A^2$ can differ considerably from σ_a^2 , and similarly the conditional variance (9.55) can vary from its unconditional value in (9.52).

In any one realization of (9.44) we effectively know $\hat{\sigma}_A^2$ from (9.58), making the conditional variance (9.55) the correct one for computing the enrichment significance level of $\mathcal{S}$. Permutation calculations, on the other hand, tend to produce the unconditional variance (9.52),

$$\text{var}_* \{\bar{z}_S^*\} \doteq \frac{1}{m} \frac{1 + (J/J_1)\sigma_a^2 C}{1 + (J/J_1)\sigma_a^2} \quad (9.60)$$

reflecting the fact that they can “see” correlations within $\mathcal{S}$ but not the effects outside of $\mathcal{S}$ that determine $\hat{\sigma}_A^2$.

The restandardization formula (9.13) depends on $m_s = 0$, sd_s from (9.58), and

$$(m_s^*, \text{sd}_s^*) \doteq (0, 1) \quad (9.61)$$

giving

$$\text{var}_{**} \{\bar{z}_S^{**}\} \doteq \frac{1 + (J/J_1)\hat{\sigma}_A^2}{1 + (J/J_1)\sigma_a^2} \text{var}_* \{\bar{z}_S^*\}. \quad (9.62)$$

To get from the permutation variance (9.60) to the conditional variance (9.55), we should multiply by

$$R \equiv \frac{1 + (J/J_1)\hat{\sigma}_A^2 C}{1 + (J/J_1)\sigma_a^2 C}. \quad (9.63)$$

Instead, the restandardization formula (9.62) multiplies (9.60) by

$$\hat{R} = \frac{1 + (J/J_1)\hat{\sigma}_A^2}{1 + (J/J_1)\sigma_a^2}. \quad (9.64)$$

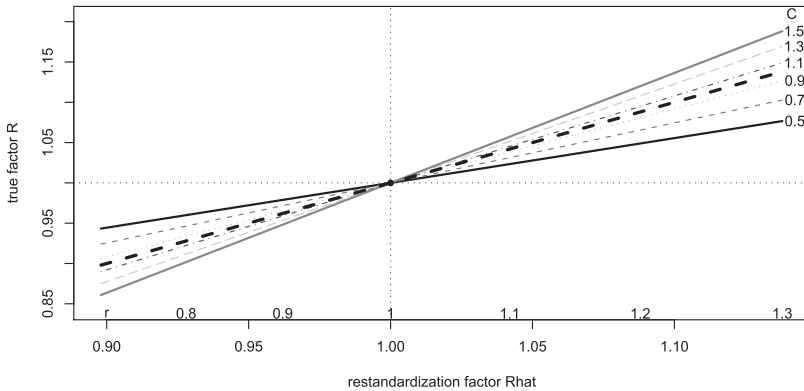


Figure 9.6 Exact variance correction factor R versus restandardization factor $\hat{R}$ (9.63)–(9.64) for values of C and $r = \hat{\sigma}_A/\sigma_a$ as labeled; $J = 5$, $\sigma_a^2 = 0.20$. $\hat{R}$ is exact for $C = 1$ (heavy dashed line).

We see that restandardization gives correct answers when $C = 1$, the average case (9.53), but is only an approximation otherwise. Figure 9.6 compares $\hat{R}$ with R for various choices of C and $r = \hat{\sigma}_A/\sigma_a$ (taking $J = 5$ and $\sigma_a^2 = 0.20$, the value from (9.48) for $\alpha = 0.10$). The restandardization results seem reasonable, though far from exact for extreme choices of C . R itself could be estimated from permutation and randomization calculations, but the special nature of our model makes this an uncertain enterprise.

Proof of Lemma 9.1 Equations (9.44) and (9.45) give

$$\sigma_0 \bar{z}_S = \bar{u}_S + \bar{v}_S \mathbf{A} \sim \mathcal{N}\left(0, \frac{1}{m} + \sigma_a^2 \|\bar{v}_S\|^2\right) \quad (9.65)$$

where $\bar{u}_S = \sum_S u_i/m$, and likewise $\bar{v}_S$ is the average of the rows of $\mathbf{V}$ indexed by S . From (9.39) and (9.50),

$$\bar{v}_S = \frac{J}{J_1} \left(\dots, \frac{m_j}{m} - \frac{1}{J}, \dots \right), \quad (9.66)$$

verifying (9.52) since

$$\|\bar{v}_S\|^2 = \left(\frac{J}{J_1}\right)^2 \sum_1^5 \left(\frac{m_j}{m} - \frac{1}{J}\right)^2 = \frac{1}{m} \frac{J}{J_1} C \quad (9.67)$$

and $\sigma_0^2 = 1 + (J/J_1)\sigma_a^2$. $\square$

Proof of Theorem 9.2 Since $\sum_j \bar{v}_{S_j} = 0$, (9.65) gives

$$\sigma_0 \bar{z}_S | \mathbf{A} \sim \mathcal{N}\left(\bar{v}_S \tilde{\mathbf{A}}, \frac{1}{m}\right) \quad (9.68)$$

where $\tilde{A}_j = A_j - \bar{A}$. But standard multivariate normal calculations show that

$$E\left\{\bar{v}_S \tilde{\mathbf{A}} \mid \hat{\sigma}_A^2\right\} = 0 \quad \text{and} \quad E\left\{\left(\bar{v}_S \tilde{\mathbf{A}}\right)^2 \mid \hat{\sigma}_A^2\right\} = \hat{\sigma}_A^2 \|\bar{v}_S\|^2 / J_1. \quad (9.69)$$

This combines with (9.67) to give (9.55), using $\sigma_0^2 = 1 + (J/J_1)\sigma_a^2$. $\square$

Exercise 9.9 Verify the preceding statement.

Non-central χ^2 calculations Equation (9.44) gives $\sigma_0 \mathbf{z} | \mathbf{A} \sim \mathcal{N}_N(\mathbf{V}\mathbf{A}, I_N)$. Applying (9.40) shows that $\|\mathbf{V}\mathbf{A}\|^2 = \delta^2$, verifying (9.56)–(9.57). Distribution $\chi_N^2(\delta^2)$ has mean and variance $(\delta^2 + N, 4\delta^2 + 2N)$, showing that the variance of $\text{sd}_s^2 | \hat{\sigma}_A^2$ is $O_p(N^{-1})$ (9.58).

Permutation calculations Model (9.36) makes the elements in any one row of $\mathbf{X}$ i.i.d. normal variates

$$x_{ik} = u_{ik} + \sum_{j=1}^J v_{ij} a_{jk} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_0^2) \quad (9.70)$$

for $k = 1, 2, \dots, n$. The permutation distribution of z_i (9.42) has mean and variance $(0, \hat{\sigma}_i^2 / \sigma_0^2)$, where $\hat{\sigma}_i^2$, the usual unbiased variance estimate $\sum_k (x_{ik} - x_i)^2 / (n - 1)$, has expectation σ_0^2 . Averaging over $i = 1, 2, \dots, N$ supports (9.61).

Let $\bar{x}_{Sk}$ be the average of x_{ik} for $i \in S$. The same argument as in (9.65)–(9.67) shows that

$$\bar{x}_{Sk} \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{1}{m} \left[1 + \frac{J}{J_1} \sigma_a^2 C\right]\right) \quad \text{for } k = 1, 2, \dots, N. \quad (9.71)$$

Also

$$\bar{z}_S = \sqrt{\frac{n}{4}} \frac{\bar{x}_{S(2)} - \bar{x}_{S(1)}}{\sigma_0}, \quad (9.72)$$

the notation indicating averages of $\bar{x}_{S_k}$ over $k = 1, 2, \dots, n/2$ and over $n/2 + 1, \dots, n$, giving approximation (9.60).

9.4 Local Averaging

Enrichment has an exotic sound in the microarray context, where the choice of gene sets seems rather mysterious, at least to those of us without the required biogenetical background. There is however one situation where gene sets are natural and enrichment a familiar tactic, though it doesn't involve genes. We will discuss the idea of *local averaging* in terms of the DTI example of Section 2.5.

The DTI data reports z -values comparing dyslexic children with normal controls at $N = 15\,443$ brain locations or *voxels*, which can be thought of as cubes roughly 1.5mm on a side. Voxel i has coordinates

$$(x_i, y_i, u_i) \quad (9.73)$$

where x_i measures units from back to front of the brain, y_i from right to left, and u_i from bottom to top. Figure 2.4 indicates the z -values on a horizontal slice of the brain about half-way from bottom to top.

Local averaging amounts to smoothing the z_i values by averaging them over nearby voxels. In the example which follows, each z_i is replaced with Z_i , the average of z_j for j in the bigger cube

$$(x_i \pm 2, y_i \pm 2, u_i \pm 2). \quad (9.74)$$

The bigger cube, call it S_i , is five voxels on a side, containing a maximum of 125 original voxels. (Only 2% of the cubes actually had 125 voxels, the average number being 81 ± 25 .)

A simple idea motivates local averaging: suppose that the voxels in S_i have independent normal z -values

$$z_j \stackrel{\text{ind}}{\sim} \mathcal{N}(\mu_j, 1) \quad \text{for } j \in S_i. \quad (9.75)$$

Then averaging results in

$$Z_i \sim \mathcal{N}(\bar{\mu}_i, 1/m_i) \quad (9.76)$$

where m_i is the number of voxels in S_i and $\bar{\mu}_i$ is the average of μ_j , $j \in S_i$. Rescaling to variance 1 gives

$$\tilde{Z}_i = \sqrt{m_i} Z_i \sim \mathcal{N}(\sqrt{m_i} \bar{\mu}_i, 1). \quad (9.77)$$

If $\bar{\mu}_i \doteq \mu_i$, then going from $z_i \sim \mathcal{N}(\mu_i, 1)$ to $\tilde{Z}_i \sim \mathcal{N}(\sqrt{m_i} \mu_i, 1)$ magnifies effect sizes in the non-null cubes.

Exercise 9.10 Suppose $\bar{\mu}_i = \mu_i$ and $m_i = m$ for all values of i . (a) How will averaging change g_1 in the hierarchical model (2.47), (2.49)? (b) If $m = 9$, $\pi_0 = 0.95$, and $g_1 \sim \mathcal{N}(0, 2)$ in (2.49), graph $\text{fdr}(z) = \pi_0 f_0(z)/f(z)$ for the original z_i 's and the $\tilde{Z}_i$'s.

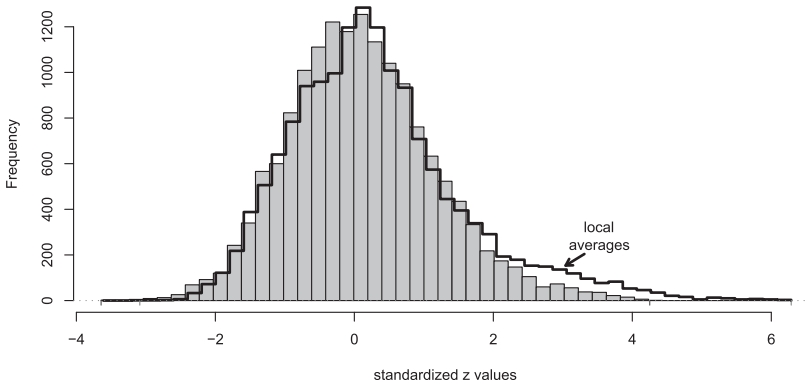


Figure 9.7a $N = 15\,443$ DTI z -values (solid histogram) compared to local averages Z_i (line histogram); standardized (9.79). Local averaging has produced a heavier right tail.

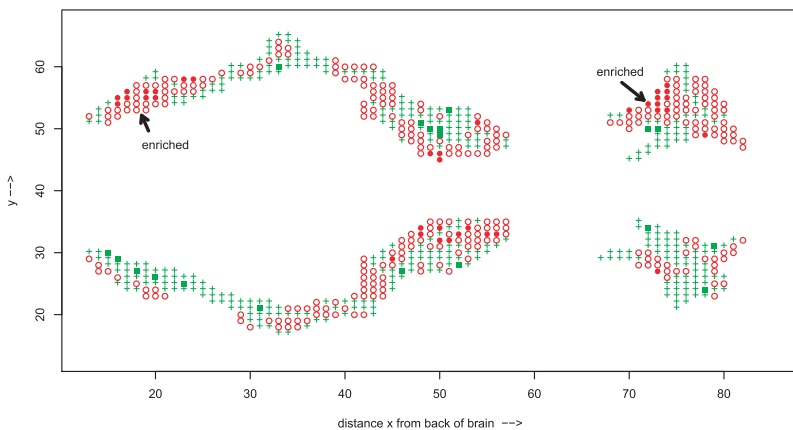


Figure 9.7b Averaged values $\tilde{Z}_i$ for DTI data (9.79); horizontal section, bottom of brain. Red, $\tilde{Z}_i > 0$; green, $\tilde{Z}_i < 0$; solid circles, $\tilde{Z}_i > 2$; solid squares, $\tilde{Z}_i < -2$. Black arrows indicate enrichment regions, $\widehat{\text{fdr}}(Z_i) \leq 0.20$.

The $N = 15\,443$ averaged values Z_i had empirical null estimates

$$(\hat{\sigma}_0, \hat{\sigma}_0, \hat{\pi}_0) = (-0.15, 0.47, 0.940) \quad (9.78)$$

using the MLE method of Section 6.3. This compares with $(-0.12, 1.06, 0.984)$ for the original z_i values pictured in Figure 5.1b. Figure 9.7a compares histograms for the standardized z -values

$$\tilde{Z}_i = (Z_i + 0.15)/0.47 \quad \text{and} \quad \tilde{z}_i = (z_i + 0.12)/1.06. \quad (9.79)$$

Averaging appears to have worked: the $\tilde{Z}$'s have a noticeably heavier right tail, though the effect is nowhere as large as suggested by (9.77). (Note that the ratio of $\hat{\sigma}_0$ values, $0.47/1.06$, is closer to $1/2$ than to the value $1/9$ suggested by $m_i \sim 81$. Positive correlation within cubes increases the variance of Z_i in (9.76).)

A local false discovery rate analysis (6.38) based on the MLE method yielded interesting regions of difference between the dyslexic brain patterns and normal controls. Figure 9.7b indicates the $\tilde{Z}_i$ values on a horizontal slice, this time at the bottom of the brain rather than the middle slice of Figure 2.4. The black triangles indicate voxels with $\widehat{\text{fdr}}(Z_i) \leq 0.20$. These “enrichment” regions, which were not found using the original z_i 's, both occur on the left side of the brain.

Local averaging is a simple form of enrichment that raises several points of comparison with the previous sections:

- The reader may wonder what has happened to restandardization (9.13)–(9.14). Our use of an empirical null in assessing $\widehat{\text{fdr}}(Z_i)$ for Figure 9.7b is in the spirit of row randomization in that it takes account of the variability of Z_j across all “gene sets” $\mathcal{S}_j$, not just within any one set of interest $\mathcal{S}_i$. Empirical null calculations can also include permutation information, as in the sixth bullet point of Section 6.5.
- There is however an important difference. Permutation calculations (9.3), (9.4) are done *separately* for each set $\mathcal{S}_i$, carrying over to restandardization (9.13)–(9.14). This takes account of set differences, e.g., different sizes m_i or correlation structures, while the $\widehat{\text{fdr}}(Z_i)$ analysis treats all sets $\mathcal{S}_i$ the same.
- On the other hand, if the experimenters had set the DTI scanner to collect data on a coarser scale, the Z_i analysis would seem natural.
- The examples of previous chapters assumed that individual cases could be treated as identical a priori. There the “gene sets” were singletons, having $m_i = 1$ and no internal correlation structure, encouraging identi-

cal treatment (though we might have Bayesian reasons for treating them differently, as in (5.21)–(5.25)).

- Suppose we did try to apply a restandardization formula to, say, the prostate data example of Section 2.1. The permutation null distribution for any one gene will be approximately $z_i^* \sim \mathcal{N}(0, 1)$, Section 6.5, making $(m_s^*, \text{sd}_s^*) \doteq (0, 1)$ in (9.13). With $s(z) = z$, we get $(m_s, \text{sd}_s) = (\bar{z}, \bar{\sigma})$, the empirical mean and standard deviation of all N z -values. Formula (9.14) then yields

$$p_i^{**} = \Phi\left(\frac{z_i - \bar{z}}{\bar{\sigma}}\right) \quad (9.80)$$

as the restandardized p -value for case i . This compares with the empirical null p -value

$$\hat{p}_i = \Phi\left(\frac{z_i - \hat{\delta}_0}{\hat{\sigma}_0}\right) \quad (9.81)$$

where $(\hat{\delta}_0, \hat{\sigma}_0)$ are the empirical null mean and standard deviation, Chapter 6.

Exercise 9.11 Verify (9.80).

The tension between row randomization and permutation brings us back to questions of “learning from the experience of others,” Section 1.4 and Section 2.6. Chapter 10 deals directly with the combination and separation of cases in a false discovery rate analysis, where results like those of Figure 9.7b will be examined more skeptically.

Notes

Gene set enrichment analysis, and the GSEA algorithm, were proposed in an influential paper by Subramanian et al. (2005) where the p53 data is one of three main examples. The Bioconductor package `limma` (Smyth, 2004) offers a GSEA-like option based on $\bar{z}_S$. Other enrichment techniques appear in Pavlidis et al. (2002) and Rahnenführer et al. (2004). Most of the material in Section 9.1 and Section 9.2, including restandardization and the maxmean statistic, is taken from Efron and Tibshirani (2007).